Axiom as Pharmakon

Disease of forgetting XOR NAND XNOR structure of knowledge.

Scoped Deductive Closure, Symmetric Adjudication, and Class-Level Uncertainty

Abstract

This note formalizes a simple relation between assumptions, proof, reuse of established results, disagreement, semantic extension, and unresolved propositions.

The central observation is that a proof does not establish an unqualified sentence independently of its formal setting. It establishes a consequence of specified assumptions under specified rules of inference and fixed interpretations of the terms occurring in the derivation.

This does not weaken proof. It is what makes proof possible.

Once a proposition has been derived, it may subsequently be reused as a premise without increasing the deductive power already present in the original assumptions. Formally, this is an immediate consequence of cut and the idempotence of deductive closure.

Conversely, a purported refutation made under the same assumptions and proof standard cannot obtain the negation of an established proposition without either:

1. exposing inconsistency or an invalid derivation;
2. introducing at least one premise not already warranted by the original assumptions; or
3. changing the interpretation or inferential rules.

Mere withholding of assent is therefore not a refutation.

When neither a proposition nor its negation is derivable, the proposition is unresolved relative to that deductive setting. In classical first-order logic, if genuine independence is established, completeness gives models of the assumptions realizing each side. Such independence is relative to the specified theory: consistent extensions can decide the previously independent sentence.

Finally, adding new vocabulary while preserving the interpretation of the old vocabulary cannot change the truth of old formulas. New distinctions may therefore extend what can be expressed without retroactively altering relations already established under the preserved interpretation.

These results provide a formal distinction between scoped knowledge, unresolved claims, changed assumptions, and changed semantics without appealing to an external or privileged adjudicating viewpoint.

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1. Formal setting

Let L be a formal language.

Let \mathrm{Sent}(L) denote its sentences.

Let

$$
\Gamma \subseteq \mathrm{Sent}(L)
$$

be a set of explicitly admitted assumptions.

Let

$$
\Gamma \vdash \varphi
$$

mean that \varphi is derivable from \Gamma under a fixed deductive calculus.

The core results require only the following structural properties.

Axiom / assumption rule

If

$$
\varphi\in\Gamma,
$$

then

$$
\Gamma\vdash\varphi.
$$

Monotonicity

If

[
\Gamma\subseteq\Delta
]

and

[
\Gamma\vdash\varphi,
]

then

[
\Delta\vdash\varphi.
]

Cut

If every premise in \Delta is derivable from \Gamma, and

[
\Delta\vdash\varphi,
]

then

[
\Gamma\vdash\varphi.
]

Plain-text form:

If Γ proves every premise used by Δ,
and Δ proves φ,
then Γ proves φ.

These conditions are deliberately weak. They capture the ordinary compositional behavior of proof.

Define the deductive closure of \Gamma by

[
\operatorname{Cn}(\Gamma)

{\varphi:\Gamma\vdash\varphi}.
]

Thus the assumptions are the generators and \operatorname{Cn}(\Gamma) is what those assumptions establish under the specified consequence relation.

---

2. Deductive Closure Theorem

Theorem 1 — Closure and reuse

For every \Gamma,

[
\Gamma\subseteq\operatorname{Cn}(\Gamma)
]

and

[
\operatorname{Cn}(\operatorname{Cn}(\Gamma))

\operatorname{Cn}(\Gamma).
]

Moreover, if

[
\Gamma\vdash\psi,
]

then

[
\operatorname{Cn}(\Gamma\cup{\psi})

\operatorname{Cn}(\Gamma).
]

Proof

The first inclusion follows immediately from the assumption rule.

For idempotence, first observe by monotonicity that

$$
\operatorname{Cn}(\Gamma)
\subseteq
\operatorname{Cn}(\operatorname{Cn}(\Gamma)).
$$

For the reverse inclusion, suppose

$$
\varphi\in
\operatorname{Cn}(\operatorname{Cn}(\Gamma)).
$$

Then

$$
\operatorname{Cn}(\Gamma)\vdash\varphi.
$$

But by definition every

$$
\theta\in\operatorname{Cn}(\Gamma)
$$

satisfies

$$
\Gamma\vdash\theta.
$$

Cut therefore gives

$$
\Gamma\vdash\varphi.
$$

Hence

$$
\varphi\in\operatorname{Cn}(\Gamma).
$$

]

Therefore

[
\operatorname{Cn}(\operatorname{Cn}(\Gamma))

\operatorname{Cn}(\Gamma).
]

Now suppose

[
\Gamma\vdash\psi.
]

Certainly

[
\Gamma
\subseteq
\Gamma\cup{\psi},
]

so monotonicity gives

[
\operatorname{Cn}(\Gamma)
\subseteq
\operatorname{Cn}(\Gamma\cup{\psi}).
]

Conversely, every premise in

[
\Gamma\cup{\psi}
]

is already derivable from \Gamma: members of \Gamma by the assumption rule and \psi by hypothesis.

Thus every derivation using

[

\Gamma\cup{\psi}
]

can be cut back to \Gamma.

Therefore

[
\operatorname{Cn}(\Gamma\cup{\psi})
\subseteq
\operatorname{Cn}(\Gamma).
]

Hence equality holds.

[
\boxed{
\operatorname{Cn}(\Gamma\cup{\psi})

\operatorname{Cn}(\Gamma)
}
]

whenever

[
\Gamma\vdash\psi.
]

∎

---

3. The exact relation between “given” and “known”

The preceding theorem gives a precise formal sense in which a proved proposition can subsequently function as a given.

Initially,

[
\Gamma
]

contains what is explicitly assumed.

After deduction,

$$
\operatorname{Cn}(\Gamma)
$$

contains what follows.

But if

$$
\psi\in\operatorname{Cn}(\Gamma),
$$

then reintroducing \psi as a premise changes nothing:

$$
\operatorname{Cn}(\Gamma\cup{\psi})

\operatorname{Cn}(\Gamma).
$$

Thus established results may be compressed and reused.

Plain-text form:

given assumptions
  -> deduction
  -> established proposition
  -> reuse that proposition as a premise
  -> no increase in deductive closure

This is not an informal convention.

It is cut.

What must not be erased is the derivational scope under which the proposition entered the closure.

---

4. Symmetric Refutation Theorem

Define consistency of \Gamma by requiring that there is no sentence \varphi for which both

$$
\Gamma\vdash\varphi
$$

and

$$
\Gamma\vdash\neg\varphi.
$$

Theorem 2 — A refutation under the same standard requires either a defect or a genuinely new premise

Assume:

$$
\Gamma\vdash\varphi
$$

and \Gamma is consistent.

Suppose some argument derives

$$
\Gamma\cup\Delta\vdash\neg\varphi.
$$

Then at least one member of \Delta is not derivable from \Gamma.

Equivalently,

$$
\Gamma\cup\Delta\vdash\neg\varphi
\quad\land\quad
\Gamma\vdash\varphi
$$

implies

$$
\exists\delta\in\Delta
$$

\text{ such that }
\Gamma\nvdash\delta,
]

unless \Gamma is inconsistent.

Proof

Assume for contradiction that every member of \Delta is derivable from \Gamma:

[
\forall\delta\in\Delta,\quad
\Gamma\vdash\delta.
]

Every member of \Gamma is also derivable from \Gamma by the assumption rule.

Hence every premise in

[
\Gamma\cup\Delta
]

is derivable from \Gamma.

But

[
\Gamma\cup\Delta\vdash\neg\varphi.
]

By cut,

[
\Gamma\vdash\neg\varphi.
]

By hypothesis,

[
\Gamma\vdash\varphi.
]

Therefore \Gamma proves both \varphi and its negation, contrary to consistency.

Hence at least one premise in \Delta was not already warranted by \Gamma.

∎

---

5. Consequence for adjudication

Theorem 2 gives an exhaustive structural classification of a purported refutation of an established result under a fixed proof standard.

If

$$
\Gamma\vdash\varphi,
$$

then a genuine challenge must do at least one of the following:

1. show that an alleged premise of \Gamma was not actually admitted;
2. show that a purported inference in the proof is invalid;
3. show that the calculus or assumptions are inconsistent;
4. change the assumptions by introducing some \delta not derivable from \Gamma;
5. change the interpretation of one or more terms;
6. change the consequence relation.

What cannot constitute a refutation under the same standard is merely:

I do not accept φ.

That sentence reports an attitude.

It does not derive

$$
\neg\varphi.
$$

Similarly,

perhaps ¬φ

does not establish

$$
\neg\varphi.
$$

If the negative claim is to be promoted under the same proof standard, it requires its own adequate derivation.

This is symmetry of standard, not symmetry of certificate shape.

An existential, a universal, and a negated universal can require different forms of evidence because their logical forms differ.

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6. Class-Level Uncertainty

There are two notions that must be kept distinct.

6.1 Operational openness

At a particular stage of inquiry, we may simply possess neither an accepted derivation of \varphi nor one of \neg\varphi.

This is a state of the proof registry.

Write:

$$
\operatorname{Open}_{\Gamma}(\varphi).
$$

It means only:

no presently accepted receipt closes φ
and
no presently accepted receipt closes ¬φ.

It does not itself prove that neither derivation exists.

6.2 Formal independence

A stronger statement is

$$
\Gamma\nvdash\varphi
$$

and

$$
\Gamma\nvdash\neg\varphi.
$$

Write:

$$
\operatorname{Indep}_{\Gamma}(\varphi).
$$

This is a metamathematical statement about the entire deductive class generated by \Gamma.

Gödel-Rosser incompleteness provides such statements for broad classes of sufficiently expressive consistent effective theories.

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7. Model-Theoretic Form of Class-Level Uncertainty

Now specialize to classical first-order logic and use its soundness and completeness theorems.

Theorem 3 — Model split for an independent sentence

Let \Gamma be a consistent first-order theory.

If

$$
\Gamma\nvdash\varphi
$$

and

$$
\Gamma\nvdash\neg\varphi,
$$

then there exist models M and N such that

$$
M\models\Gamma,
\qquad
M\models\varphi,
$$

while

$$
N\models\Gamma,
\qquad
N\models\neg\varphi.
$$

Proof

Suppose there were no model of

$$
\Gamma\cup{\varphi}.
$$

Then

$$
\Gamma\models\neg\varphi.
$$

By completeness,

$$
\Gamma\vdash\neg\varphi,
$$

contrary to hypothesis.

Therefore there exists a model M with

$$
M\models\Gamma\cup{\varphi}.
$$

Similarly, if no model satisfied

$$
\Gamma\cup{\neg\varphi},
$$

then

$$
\Gamma\models\varphi
$$

and completeness would yield

$$
\Gamma\vdash\varphi,
$$

again contradicting the hypothesis.

Therefore there exists N satisfying

$$
\Gamma\cup{\neg\varphi}.
$$

∎

This is not a third truth value.

It is a class-level fact about what the fixed assumptions determine.

---

8. Undecidability is relative to the specified deductive setting

Corollary 3.1 — Independence does not mean “undecidable forever”

If

$$
\operatorname{Indep}_{\Gamma}(\varphi),
$$

then both

[
\Gamma+\varphi
]

and

[
\Gamma+\neg\varphi
]

are consistent, assuming classical first-order logic and consistency of \Gamma.

Moreover,

[
\Gamma+\varphi\vdash\varphi
]

and

[
\Gamma+\neg\varphi\vdash\neg\varphi.
]

Thus a sentence undecidable in \Gamma becomes decided in either extension.

Proof

Consistency follows from Theorem 3.

Derivability follows immediately from the assumption rule.

∎

Therefore:

undecidable in Γ

does not entail:

undecidable under every possible extension or reinterpretation.

The latter is a different universal statement with a different quantified domain and requires a separate proof.

Gödel's theorem does not need to be weakened to make this point.

Its own statement already has a formally specified jurisdiction.

---

9. Syntax is not semantics

Let L be a language containing symbols.

A symbol is syntactic.

Its denotation or relational role is supplied by an interpretation.

Thus the same mark may occur in two structures with different meanings.

Formal semantics therefore distinguishes:

[
\text{symbol}
]

from

[
\text{interpretation of symbol}.
]

A proof requires sufficiently fixed interpretation to determine what its formulas assert.

This does not imply that the symbol has one immutable meaning under every future use.

---

10. Semantic Stability Theorem

Let

[
L\subseteq L'
]

be languages.

Let M be an L-structure.

Let M' be an L'-structure whose reduct to L is exactly M.

That means every symbol already belonging to L retains exactly its previous interpretation.

Theorem 4 — Adding new vocabulary while preserving old meanings cannot alter old truth

For every L-formula \varphi and every suitable assignment s,

[
M\models\varphi[s]
]

if and only if

[
M'\models\varphi[s].
]

Proof

Proceed by structural induction on \varphi.

Atomic formulas

All symbols occurring in an L-atomic formula belong to L.

By assumption, M' interprets every L-symbol exactly as M does.

Therefore the atomic formula has the same truth value in both structures.

Negation

Suppose the result holds for \psi.

Then

[
M\models\neg\psi
]

iff

[
M\not\models\psi.
]

By the induction hypothesis this is equivalent to

[
M'\not\models\psi,
]

which is equivalent to

[
M'\models\neg\psi.
]

Boolean connectives

The argument is identical componentwise for conjunction, disjunction, implication, and the other truth-functional connectives.

Quantifiers

Suppose the result holds for \psi(x).

The underlying L-domain is unchanged by the expansion.

Therefore

[
M\models\forall x,\psi(x)
]

iff every assignment differing only in x satisfies \psi in M.

By induction this occurs iff every such assignment satisfies \psi in M'.

Hence

[
M'\models\forall x,\psi(x).

]

Existential quantification follows similarly.

Therefore every old formula retains its truth value.

∎

This theorem gives a precise form to the statement:

adding a new distinction does not alter an old relation
if the interpretation of the old terms is actually preserved.

New vocabulary can expose distinctions that were previously inexpressible.

It cannot retroactively modify old formulas while simultaneously claiming to have left their meanings fixed.

If an old symbol is reinterpreted, that is not a meaning-preserving expansion. It is a changed interpretation and must be identified as such.

---

11. Isomorphism and transport between independently described structures

Let M and N be L-structures and let

[
h:M\cong N
]

be an isomorphism.

Theorem 5 — Isomorphic structures preserve first-order relations

For every L-formula \varphi and assignment s,

[
M\models\varphi[s]
]

if and only if

[

N\models\varphi[h\circ s].
]

Proof

By induction on terms and formulas.

The atomic case follows because an isomorphism preserves the interpretations of constants, functions, and relations.

The Boolean cases follow immediately from induction.

The quantifier cases follow because h is bijective, so ranging over the domain of M corresponds exactly to ranging over the domain of N.

∎

Thus the relation between two independently presented structures does not have to be taken as a primitive given.

It can itself be proved.

Once proved, relevant consequences can be transported.

---

12. Self-application of an equal proof standard

We can now state the central result.

Theorem 6 — Symmetric Adjudication Theorem

Fix:

1. a language L;
2. explicit assumptions \Gamma;
3. a consequence relation \vdash satisfying the structural rules above;
4. fixed interpretations for the terms used in the claim.

Suppose a conclusion \varphi is accepted exactly when an adequate derivation establishes

[
\Gamma\vdash\varphi.
]

Then any objection seeking to establish \neg\varphi under the same standard is itself subject to the same consequence relation.

If

$$
\Gamma\vdash\varphi
$$

and \Gamma is consistent, then an objection can overturn the result only by:

1. invalidating a premise;
2. invalidating an inference;
3. exposing inconsistency;
4. introducing a new premise not derivable from \Gamma;
5. changing the interpretation;
6. or changing the proof standard.

Mere non-acceptance cannot constitute a derivation of \neg\varphi.

Proof

If the objection uses no additional premises and the same consequence relation, then a successful objection would establish

$$
\Gamma\vdash\neg\varphi.
$$

Together with

$$
\Gamma\vdash\varphi,
$$

this contradicts consistency.

If the objection uses additional premises \Delta, Theorem 2 proves that at least one of them is not already derivable from \Gamma, unless the original assumptions are inconsistent.

If an interpretation is changed, the argument concerns a different interpreted structure unless an explicit preservation theorem transports the result.

If the inference rules are changed, the argument is no longer made under the original consequence relation.

These cases exhaust the ways the formal setting can change.

Therefore personal non-acceptance supplies no additional logical possibility.

∎

---

13. Why this theorem is not self-sealing

The theorem explicitly permits its own defeat.

A critic may show:

a premise is false;
a premise was never actually admitted;
a derivation step is invalid;
the supposed consequence relation lacks cut or another stated property;
the interpretation was incorrectly specified;
an alleged isomorphism fails;
the theory is inconsistent;
or the theorem has exceeded its declared scope.

Any one of these is a genuine mathematical objection.

What is excluded is only the inference:

I have supplied no defect and no counter-derivation,
but I remain unconvinced;
therefore the proposition has been refuted.

That does not have the logical form of a refutation.

---

14. The formal analogue of the “privileged viewpoint” error

Suppose a proof establishes

[
\Gamma\vdash\varphi.

]

That does not by itself establish

[
\text{“}\varphi\text{ under every possible extension,
interpretation, language, and future distinction.”}
]

The second statement has a larger quantifier domain.

To prove it, that domain must itself be specified and covered by an appropriate theorem.

Silently moving from

all models / interpretations covered by this proof

to

all possible meanings whatsoever

is a quantifier-scope change.

It is not licensed by the original derivation.

Equally, however, the possibility of another interpretation does not refute what was proved under the original one.

Both errors are prevented by keeping the scope explicit.

---

15. Class-level closure by an empty solution set

Many familiar theorems have exactly the form under discussion.

Let C be a defined class and let R(x) be a fixed relation.

Define its solution set

[
Z_R

{x\in C:R(x)}.

]

A universal negative theorem

[
\forall x\in C,\neg R(x)
]

is equivalent to

[
Z_R=\varnothing.
]

This is a class-level closure.

It does not require enumeration of every member of C.

---

16. Polygon example

For an inscribed regular n-gon in a unit circle,

[
P_n

2n\sin(\pi/n).
]

For every finite integer

[
n\ge3,
]

we have

[
0<\pi/n
]

and the standard inequality

[
\sin x<x
]

for positive x.

Hence

[
2n\sin(\pi/n)
<
2n(\pi/n)

2\pi.
]

Therefore

[
\forall n\in\mathbb N,\ n\ge3
\Rightarrow
P_n\ne2\pi.
]

Equivalently,

[
{n\ge3:P_n=2\pi}

\varnothing.
]

The polygons converge to the circle in perimeter:

[
P_n\to2\pi,
]

but no finite polygon in the specified class closes the equality.

This is proved class-wide by a relation, not by enumerating polygons.

---

17. Exact proportion as an integer relation

A clean rational proportion between a quantity x and 1 exists exactly when there are integers a,b, with a\ne0, satisfying

[
ax=b.
]

Thus

[
x\in\mathbb Q
]

iff

[
\exists a,b\in\mathbb Z,\quad
a\ne0
\quad\land\quad
ax-b=0.
]

Therefore irrationality of \pi can be written as the class-level null statement

[
\forall a,b\in\mathbb Z,\quad
a\ne0
\Rightarrow
a\pi-b\ne0.
]

Equivalently,

[
{
(a,b)\in\mathbb Z^2:
a\ne0\land a\pi=b
}

\varnothing.
]

No finite search over integer ratios establishes this.

A proof of irrationality closes the entire defined class at once.

---

18. Transcendence as stronger class-level closure

A number x is algebraic precisely when there exists a nonzero polynomial

$$
p(X)\in\mathbb Z[X]
$$

such that

$$
p(x)=0.
$$

Lindemann's transcendence theorem for \pi therefore establishes

$$
\forall p\in\mathbb Z[X]\setminus{0},
\quad
p(\pi)\ne0.
$$

Equivalently,

$$
{
p\in\mathbb Z[X]\setminus{0}:
p(\pi)=0
}

\varnothing.
$$

This enlarges the closed class:

no nontrivial linear integer relation closes on π

becomes

no nontrivial finite polynomial integer relation closes on π.

Again, the proof does not reside in inspecting the decimal expansion of \pi.

Nor does it reside in testing polynomial after polynomial.

The theorem closes the defined relation class.

Once established, the result may be reused as a premise. By Theorem 1, such reuse is deductively conservative relative to assumptions that already prove it.

---

19. Gödel as the complementary form

Lindemann supplies a class-level closure:

[
\forall p\ne0,\quad p(\pi)\ne0.
]

Gödel-Rosser supplies, under its stated hypotheses on a formal theory T, a different kind of class-level theorem:

there exists a sentence G_T such that

[
T\nvdash G_T
]

and

[
T\nvdash\neg G_T.
]

The theorem does not say that G_T is “unknowable forever.”

It establishes a precise limitation of derivability from T.

A suitable extension may decide the sentence.

What survives recursively is not a magical object called uncertainty.

What survives is the requirement that every promoted conclusion be warranted relative to explicitly stated assumptions, semantics, and inferential rules.

---

20. Application to the Rule 30 center problem

Let Q(t) denote the center-cell sequence generated by the stipulated Rule 30 construction.

Define the set of candidate eventual-period certificates

[
E_Q

\left{
(P,T):
P>0
\ \land
\forall t\ge T,;
Q(t+P)=Q(t)
\right}.
]

Then the alternatives become exact set-theoretic statements.

A — eventual periodicity

[
E_Q\ne\varnothing.
]

Expanded:

[
\exists P>0;
\exists T;
\forall t\ge T,;
Q(t+P)=Q(t).
]

B — non-eventual-periodicity

[

$$
E_Q=\varnothing.
$$

Expanded:

$$
\forall P>0;
\forall T;
\exists t\ge T,;
Q(t+P)\ne Q(t).
$$

C — current unresolved status

No accepted derivation presently closes either

$$
E_Q\ne\varnothing
$$

or

$$
E_Q=\varnothing.
$$

C is not a third truth value.

It is the status of the adjudication.

A proof of A need not enumerate the infinite tail. It may supply a finite structural theorem that handles arbitrary $t\ge T$.

A proof of B need not enumerate all candidate periods. It may supply a finite structural theorem that, for arbitrary P,T, produces or guarantees a violating t.

Thus:

A requires class-level positive closure.

B requires class-level negative closure.

Finite computation may provide witnesses against particular candidates
without thereby closing the quantified class.

C remains until one of those quantified relations is closed.

---

21. The general theorem schema

The entire argument can be compressed to the following schema.

Let

$$
(\Gamma,\vdash,\mathcal I)
$$

specify:

- explicit assumptions Γ;
- an inferential relation $\vdash$;
- a fixed interpretation $\mathcal I$ for the terms currently used.

Then:

1. Derivation

$$
\Gamma\vdash\varphi
$$

licenses φ under that setting.

2. Reuse

If

$$
\Gamma\vdash\varphi,
$$

then adding φ as a premise does not enlarge deductive closure.

3. Symmetry

A claim of $\neg\varphi$ made under the same standard requires its own valid derivation.

4. Changed premises

If a negative derivation requires a premise not derivable from \Gamma, then the argument has changed its assumptions.

5. Changed semantics

If old terms are reinterpreted, the argument has changed its interpreted structure.

6. Preserved semantic extension

If new distinctions are added while old meanings remain fixed, all old formulas retain their prior truth values.

7. Unresolved status

If neither side has an accepted closure, neither side may be promoted solely because the other remains unproved.

8. Self-application

An objection presented under this same proof standard is itself a claim and is subject to the same requirements.

There is no additional logical category in which an adjudicator's unsupported rejection becomes equivalent to a proof.

---

22. Plain-language statement

A proof begins with things explicitly taken as given.

It applies relations whose use has been specified.

It ends with something that follows from those givens.

Once that result has been proved, it can be reused without proving it from scratch each time. That is ordinary deductive compression.

But the compression does not erase where the result came from.

If another person wants to negate the result under the same assumptions and rules, their negation has to meet the same standard.

They may find a mistake.

They may reject an assumption.

They may add a new assumption.

They may reinterpret a term.

They may introduce a stronger language.

They may prove that two representations are or are not structurally equivalent.

All of those are legitimate mathematical moves.

What they cannot do under one equal proof standard is convert

“I have not accepted your proof”

into

“I have proved its negation.”

Likewise, a theorem established under fixed meanings does not become false merely because later language introduces a finer distinction.

If the old meanings are retained, the old relation remains.

The finer language may then support new propositions that could not previously be expressed.

This is extension, not retroactive contradiction.

---

23. Lean formalization target

The deductive core requires very little machinery.

A Lean development can begin from an abstract consequence relation with three structural laws:

Prov Γ φ

assumption:
  φ ∈ Γ -> Prov Γ φ

monotonicity:
  Γ ⊆ Δ -> Prov Γ φ -> Prov Δ φ

cut:
  (forall ψ ∈ Δ, Prov Γ ψ) ->
  Prov Δ φ ->
  Prov Γ φ

Define:

Cn Γ := {φ | Prov Γ φ}

The first machine targets are then:

theorem closure_idempotent :
  Cn (Cn Γ) = Cn Γ

theorem proved_as_premise_is_conservative
  (h : Prov Γ φ) :
  Cn (insert φ Γ) = Cn Γ

For negation:

Consistent Γ :=
  forall φ, not (Prov Γ φ and Prov Γ (neg φ))

and the central refutation theorem becomes:

theorem negative_derivation_requires_new_premise
  (hpos : Prov Γ φ)
  (hcons : Consistent Γ)
  (hneg : Prov (Γ ∪ Δ) (neg φ)) :
  exists δ, δ ∈ Δ and not (Prov Γ δ)

The semantic-extension theorem can then be formalized using first-order structures:

L ⊆ L'
M' is an expansion of M
---
for every old-language formula φ,

Sat M φ iff Sat M' φ

and the isomorphism theorem:

h : M ≃□ N

Sat M φ s iff Sat N φ (h ∘ s)

These are ordinary structural-induction proofs.

The Rule 30 endpoint can be represented separately by defining Q, E_Q, and the formulas corresponding to

E_Q ≠ ∅

and

E_Q = ∅.

No special philosophical primitive is required in the Lean development.

The philosophical vocabulary belongs to the interpretation of these proved structural facts, not to their formal validity.

Conclusion

The fundamental formal relation is:

[
\boxed{
\text{explicit givens}
\longrightarrow
\text{deductive closure}
}
]

with

[
\boxed{
\operatorname{Cn}(\operatorname{Cn}(\Gamma))

\operatorname{Cn}(\Gamma)
}
]

and therefore

[
\boxed{
\Gamma\vdash\varphi
\Rightarrow
\operatorname{Cn}(\Gamma\cup{\varphi})

\operatorname{Cn}(\Gamma).
}
]

This explains why established knowledge can subsequently function as a given without being re-proved.

Consistency then supplies the reciprocal constraint:

[
\boxed{
\Gamma\vdash\varphi
\Rightarrow
\Gamma\nvdash\neg\varphi
}
]

for a consistent \Gamma.

And cut gives the stronger adversarial result:

[
\boxed{
\Gamma\vdash\varphi
\land
\Gamma\cup\Delta\vdash\neg\varphi
\Rightarrow
\text{some premise in }\Delta
\text{ is genuinely new}
}
]

unless the original setting is inconsistent or a derivation is invalid.

Finally, semantic expansion gives:

[
\boxed{
\text{new distinctions}
+
\text{preserved old interpretations}
\Rightarrow
\text{preserved old truths}.
}
]

These statements jointly establish the desired class-level distinction:

what has been proved under fixed relations may be reused as established;

what has not been closed remains unresolved;

a change of premises, semantics, or proof rules must be made explicit;

and an objection is subject to the same proof standard as the claim it attacks.

No privileged adjudicating viewpoint is required.

++++++++++

```
-- Abstract consequence relation with the three structural laws
class Consequence (Sentence : Type) where
  Prov : Set Sentence → Sentence → Prop
  assumption : ∀ {Γ φ}, φ ∈ Γ → Prov Γ φ
  monotonicity : ∀ {Γ Δ φ}, Γ ⊆ Δ → Prov Γ φ → Prov Δ φ
  cut : ∀ {Γ Δ φ},
    (∀ ψ ∈ Δ, Prov Γ ψ) → Prov Δ φ → Prov Γ φ

namespace Consequence
variable {Sentence : Type} [Consequence Sentence]
open Consequence

def Cn (Γ : Set Sentence) : Set Sentence :=
  {φ | Prov Γ φ}

theorem closure_idempotent (Γ : Set Sentence) :
```

```
    Cn (Cn Γ) = Cn Γ := by
  apply Set.ext
  intro φ
  constructor
  · -- Cn (Cn Γ) ⊆ Cn Γ
    intro h
    have : Prov (Cn Γ) φ := h
    have hΔ : ∀ ψ ∈ Cn Γ, Prov Γ ψ := fun ψ hψ => hψ
    exact cut hΔ this
  · -- Cn Γ ⊆ Cn (Cn Γ)
    intro h
    exact monotonicity (fun _ hx => hx) h

theorem proved_as_premise_is_conservative
    {Γ : Set Sentence} {φ : Sentence}
    (h : Prov Γ φ) :
    Cn (insert φ Γ) = Cn Γ := by
  apply Set.ext
  intro ψ
  constructor
  · intro hψ
    -- every premise of Γ ∪ {φ} is already provable from Γ
    have hΔ : ∀ θ ∈ insert φ Γ, Prov Γ θ := by
      intro θ hθ
      cases hθ with
      | inl hφ => rw [hφ]; exact h
      | inr hΓ => exact assumption hΓ
    exact cut hΔ hψ
  · intro hψ
    exact monotonicity (Set.subset_insert _ _) hψ

def Consistent (Γ : Set Sentence) : Prop :=
  ∀ φ, ¬ (Prov Γ φ ∧ Prov Γ (neg φ))   -- assume a negation operation

theorem negative_derivation_requires_new_premise
    {Γ Δ : Set Sentence} {φ : Sentence}
    (hpos  : Prov Γ φ)
    (hcons : Consistent Γ)
    (hneg  : Prov (Γ ∪ Δ) (neg φ)) :
    ∃ δ ∈ Δ, ¬ Prov Γ δ := by
  by_contra h
  push_neg at h
  -- every member of Δ is provable from Γ
  have hΔ : ∀ δ ∈ Δ, Prov Γ δ := fun δ hδ => (h δ).2 hδ
```

```
  -- therefore every premise of Γ ∪ Δ is provable from Γ
  have hAll : ∀ θ ∈ Γ ∪ Δ, Prov Γ θ := by
    intro θ hθ
    cases hθ with
    | inl hΓ => exact assumption hΓ
    | inr hΔ' => exact hΔ θ hΔ'
  -- cut yields Prov Γ (neg φ)
  have : Prov Γ (neg φ) := cut hAll hneg
  -- contradiction with consistency
  exact hcons φ ⟨hpos, this⟩

end Consequence
```